Symmetry-Optimized Image Guidance and Biomechanics-Driven Force Measurement for Robotic Ultrasound: A Control Framework Enhancing Sonographic Quality

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***Abstract*—Robotic ultrasound systems (RUSS) face significant challenges in maintaining optimal image quality and contact force across complex, deformable anatomical structures. This article proposes a control framework integrating symmetry-optimized image guidance with biomechanics-driven force measurement. We introduce a column similarity (CS) method for rapid in-plane tracking and a Young–Laplace (Y–L) model to predict surface curvature changes for out-of-plane adaptation. To ensure clinical safety during autonomous operation, a dual-layer protection mechanism combining global force thresholds with adaptive image-gain limits is integrated. To validate the efficiency of the proposed visual algorithm, the in-plane module was tested on an embedded platform (NVIDIA Jetson), achieving a processing rate of 66 Hz. Extensive in vivo experiments were conducted on highly heterogeneous anatomical structures, including the curvature-rich supraclavicular region and the deformable abdominal aorta. Results demonstrate that the proposed framework elevates a composite image quality score (IQS) by 27.2% compared to conventional methods, with the gap between the proposed method and human expert within 5%. Furthermore, the autonomous system completes in vivo abdominal aorta volumetric scans in 49 s, achieving vessel reconstruction quality comparable to manual scanning by expert sonographers (dice similarity coefficient (DSC) 0.85 and average symmetric surface distance (ASSD) 2.9 mm). The findings confirm the system's capability for high-precision, autonomous sonography in diverse clinical scenarios.**

***Index Terms*—Autonomous ultrasound scanning, coupled stability, force control, medical robotics, ultrasound imaging quality.**

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I. Introduction

AUTONOMOUS ultrasound imaging system has become a promising solution for early disease screening, especially in the absence of skilled ultrasound technicians [1]. Robotic systems are integrated with ultrasound transducers and multimodal sensors (e.g., force-torque sensor) to localize internal lesions and reconstruct 3-D biomechanical volume of potential pathological regions, enabling quantifiable substrate characterization for precision diagnostics. The overall architecture and representative autonomous scanning workflow of the proposed system are illustrated in Fig. 1. Compared to conventional imaging techniques such as X-ray or computed tomography (CT), ultrasound offers a noninvasive, radiation-free, and cost-effective alternative [2], [3]. However, robot-assisted ultrasound reconstruction often suffers from suboptimal imaging quality, as precise control of both the contact force and the angle between the probe and the epidermal surface normal is critical for high-fidelity ultrasound imaging. Even for experienced operators, maintaining consistent probe alignment and contact force can be challenging [4]. As a result, integrating multisensor robotic ultrasound systems (RUSSs) capable of tracking surface normals while suppressing artifacts and mitigating acoustic coupling deficiencies has emerged as a critical frontier in ultrasound imaging system research.

Various control strategies have been proposed to address this challenge, many of which leverage high-precision force sensors and robotic arms. Early studies by Pierrot et al. [5] and Conti et al. [6] combined force sensors with ultrasound imaging to reconstruct internal body structures. Subsequent works, including those by Chatelain et al. [7] and Virga et al. [8], investigated the relationship between contact force magnitude and image quality. Burcher et al. [9] developed a hybrid system that utilized both force sensor feedback and positional data for correction. Similarly, Wang et al. [10] proposed a contact model to mitigate the effects of tissue deformation caused

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by probe movement. Further advances by Wang et al. [11] introduced a six-degree-of-freedom (6-DoF) hybrid control scheme to improve the stability of probe-patient interactions, while Ma et al. [12] designed a mechanical system equipped with sensors to measure the distance between the probe and the skin. Additional studies by Bao et al. [13] and Fu et al. [14] focused on force control strategies to enhance operational safety, efficiency, and patient comfort. Despite these advances, many of these approaches rely solely on force feedback, leading to inconsistent imaging quality across different anatomical regions due to the lack of real-time ultrasound image feedback.

In addition to force control, accurate alignment of the ultrasound probe with the epidermal surface normal has been shown to significantly improve imaging quality. The information provided by ultrasound images can help identify in-plane normals, a process referred to as in-plane motion. Boezaart and Ihnatsenka [15] demonstrated that optimal ultrasound image quality is achieved when the probe is axially aligned with the surface normal. Most existing in-plane servo control methods rely on ultrasound confidence map (CM) algorithms, first introduced by Karamalis et al. [16]. This technique, which simulates sound wave propagation through biological tissues and assigns confidence scores to each pixel, has been widely adopted in autonomous scanning systems [10], [17], [18], [19], [20]. Enhancements to in-plane control have also been achieved by integrating image moments [21] or using force/torque sensors for normal estimation [22]. For instance, Raina et al. [23] employed Bayesian optimization for in-plane control. In contrast, the force-based method by Jiang et al. [22], while accurate in static situations, relies on a fan-like exploratory motion. This approach is incompatible with continuous scanning, resulting in an inefficient "stop-and-go" procedure, and its underlying mechanical model fails under dynamic forces like friction. Improving the robustness and speed of in-plane control methods remains a key challenge in autonomous ultrasound scanning research.

Out-of-plane motion, which involves probe movement perpendicular to the ultrasound image plane and the epidermal surface, is another critical factor in achieving comprehensive scanning coverage. Unlike in-plane motion, out-of-plane adjustments lack direct feedback, requiring integration of in-plane image data or force/torque sensor readings. Jiang et al. [22] estimated out-of-plane normals using a force/torque sensor, but the method was sensitive to probe movement. To improve stability, Welleweerd et al. [24] analyzed probe movement to estimate out-of-plane normals in phantom experiments. Recent advancements have explored machine learning approaches to integrate multimodal sensor data for out-of-plane motion control [25], [26]. For example, Duan et al. [26] introduced a machine learning framework incorporating external force constraints, while Yang et al. [27] and Huang et al. [28] demonstrated autonomous ultrasound scanning in the spine and carotid artery. However, many of these methods fail to account for the geometric characteristics of the scanned surface, which is critical for accurate out-of-plane motion control. Consequently, further research is needed to enhance the robustness and accuracy of out-of-plane servo mechanisms.

This study aims to achieve high-quality, fully autonomous robotic 3-D ultrasound volume acquisition on the human body. We propose a novel scanning framework that relies solely on ultrasound imaging and contact force feedback, eliminating the need for complex stereo vision calibration and the probability of error accumulation [29]. To address these issues, we propose a hybrid control framework. For in-plane adjustments, we utilize a column similarity (CS) method to ensure rapid feature tracking. For out-of-plane adaptation, we employ a Young–Laplace (Y–L) model, which efficiently estimates surface curvature from pressure distribution under quasi-static conditions, avoiding the computational complexity of finite element modeling. Experimental results demonstrate the effectiveness of this framework in autonomous scanning of phantom surfaces without prior surface information. On phantom models, the system achieves a surface reconstruction accuracy of 4.9 mm/7.8° and completes a phantom scan in 259 s, a significant improvement in efficiency that meets clinical time requirements. More importantly, its clinical viability has been validated on complex human anatomies. Beyond the initial tests on the proximal humerus, the system successfully performed high-precision reconstructions of highly curved regions like the clavicle and ribs, as well as deep structures such as the abdominal aorta. In the latter task, the system demonstrated high clinical agreement with expert scans, achieving a dice similarity coefficient (DSC) of 0.85.

1) A lightweight in-plane servo control algorithm based on CS, offering superior control bandwidth and robustness compared to conventional methods. The algorithm was implemented on an embedded platform (NVIDIA Jetson), demonstrating real-time processing capabilities (66 Hz).
2) An out-of-plane servo control method grounded in biomechanics, leveraging the Y–L equation for superior imaging quality on irregular surfaces.
3) A comprehensive autonomous scanning strategy that predicts epidermal surfaces to determine optimal scanning orientations, facilitating high-quality coverage of unknown surfaces and human subjects.
4) A series of in vivo experiments on diverse anatomical structures (e.g., aorta and clavicle) verify that the autonomous system achieves reconstruction quality comparable to expert sonographers.

II. System Overview and Scanning Strategy

To achieve autonomous US scanning, it is necessary to describe the scanning process and establish the framework of the scanning system. This section starts with an evaluation of factors affecting probe motion and image quality. Next, a physical model is developed to describe the scanning task. Finally, the control framework proposed in this study is presented.

A. Image Quality With Different Probe Direction

To rigorously quantify the impact of probe orientation on sonographic outcomes, we first establish a unified evaluation standard, the image quality score (IQS). This composite metric

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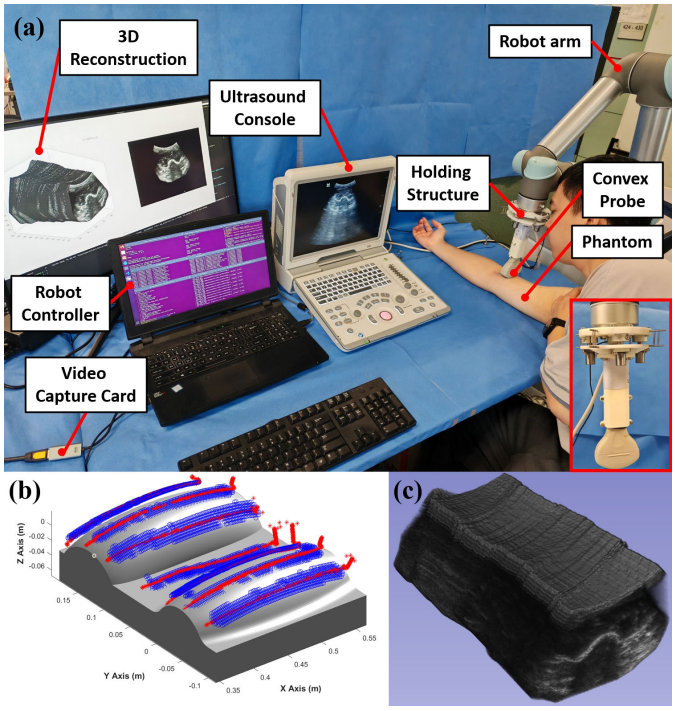


Fig. 1. Overview of the autonomous scanning system, including (a) system overview, (b) phantom scanning and surface estimation, and (c) autonomous reconstruction of the upper arm 3-D volume model.

integrates the zeroth-order moment of the US CM (I_{conf}) [16]—which assesses structural integrity—with three standard clinical indicators: signal-to-noise ratio (SNR), relative contrast (C_{rel}), and contrast-to-noise ratio (CNR). They are formulated as

$$I_{conf} = \iint_{\Omega} C(x,y)\,dxdy, \quad \text{SNR} = \frac{\mu_{sig}}{\sigma_{noise}} \tag{1}$$

$$C_{rel} = \frac{\mu_{sig} - \mu_{bg}}{\mu_{bg}}, \quad \text{CNR} = \frac{|\mu_{sig} - \mu_{bg}|}{\sigma_{noise}} \tag{2}$$

where μ_{sig} and μ_{bg} denote the mean intensities of the signal and background regions of interest (ROI), respectively. The final IQS is the average of these four metrics after min-max normalization, defined as follows:

$$\text{IQS} = \frac{1}{4}\sum_{i=1}^{4} \frac{M_i - \min(M_i)}{\max(M_i) - \min(M_i)} \tag{3}$$

where M_i represents the measurement value for the ith index, and $M_{\max}$ and $M_{\min}$ denote the maximum and minimum value among the set.

Equal weights (1/4) are chosen for control stability: in a closed-loop robotic system, dynamically adapting weights based on real-time noise levels would produce a time-varying objective function, potentially causing motion oscillations. Since all four submetrics are normalized to $[0, 1]$, uniform weighting yields an unbiased composite score that remains sensitive to both structural degradation (via I_{conf}) and acoustic degradation (via SNR and CNR) simultaneously. This ensures the controller responds consistently regardless of the dominant degradation mode.

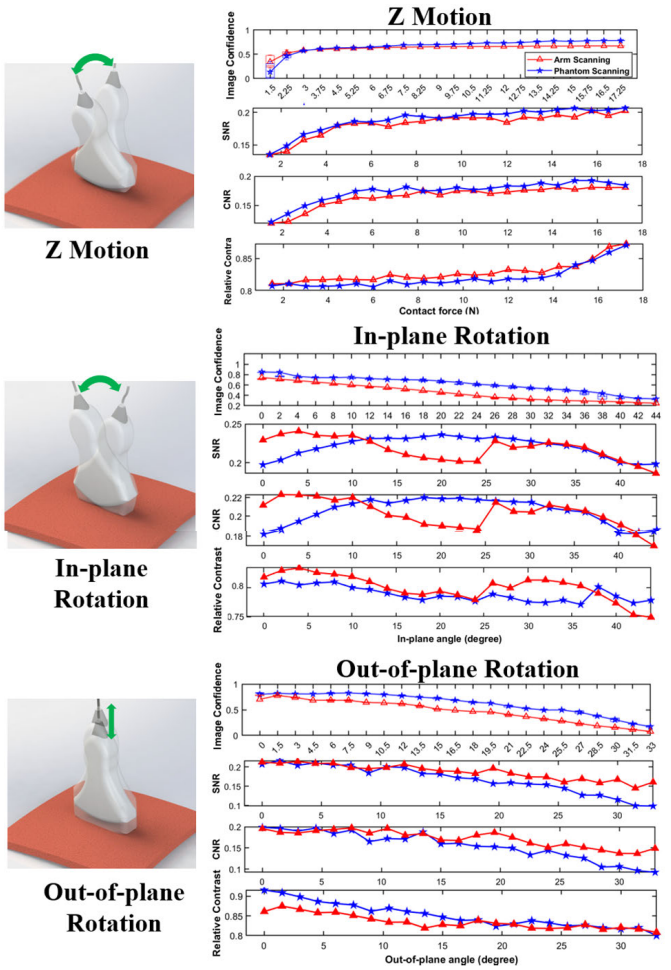


Fig. 2. Impact of various probe motion on image quality in phantom and volunteers' body, including out-of-plane motion, in-plane motion, and Z-axis motion.

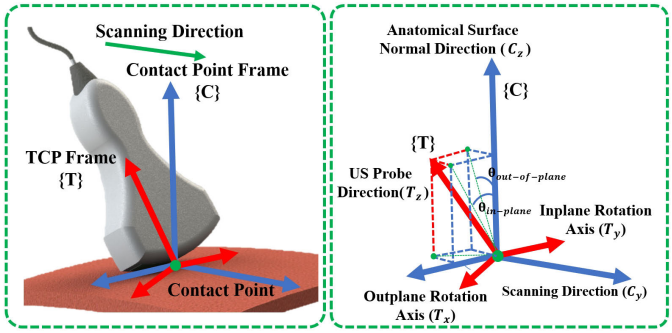


Fig. 3. Ultrasound probe-skin contact model.

Sensitivity Analysis and Metric Complementarity: Analysis against probe motion (Fig. 2) shows that contact force and out-of-plane tilt affect all four metrics consistently, confirming their dependence on acoustic coupling. However, in-plane rotation ($\theta_{in\text{-}plane}$) reveals a divergence: standard metrics (SNR, C_{rel}, and CNR) remain insensitive, whereas I_{conf} peaks sharply at optimal alignment. This occurs because in-plane rotation affects the *geometric centering* of the anatomy rather than signal strength. This justifies our composite IQS: relying solely on SNR or CNR fails to guide in-plane alignment.

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Consequently, our framework uses I_{conf} as the primary feedback for in-plane servoing, utilizing the full IQS for global validation.

Regarding contact force, the IQS plateau (e.g., ~9N) does not imply a universal optimal force but highlights the limitation of fixed-force strategies during dynamic scanning over varying tissue. Therefore, our control framework aims to maintain surface normality while *actively modulating* contact force to sustain a consistently high IQS, rather than targeting a static force value.

B. System Model and Problem Description

In this study, Cartesian coordinates are used to define the position and orientation of the US probe and contact point, as shown in Fig. 3. Within the tool center point (TCP) frame $\{\mathbf{T}\}$, which is fixed to the US probe, T_z represents the axial direction of the probe, while T_y indicates the direction normal to the US image plane. Similarly, the frame $\{\mathbf{C}\}$ is used to describe the pose of the contact point, where C_z corresponds to the normal direction of the anatomical surface at the contact point, and C_y denotes the scanning direction. To achieve optimal imaging quality, it is crucial that T_z aligns with C_z. Consequently, the control framework developed in this research endeavors to regulate the robotic arm such that $\theta_{in\text{-}plane}$ and $\theta_{out\text{-}of\text{-}plane}$ are maintained near zero, while simultaneously estimating C_y to guide the scanning trajectory of the probe.

In conclusion, this study concentrates on two aspects of probe motion. The first aspect aims to enhance image quality and involves rotation around T_x, T_y, and translation along T_z. The second aspect is to achieve comprehensive scanning coverage, which involves rotation around T_z and translation along T_y. Section II-C will concentrate on controlling these probe movements.

C. Scanning Strategy and Control Framework

The proposed system regulates Z-motion, $\theta_{in\text{-}plane}$, and $\theta_{out\text{-}of\text{-}plane}$ to autonomously track the epidermal surface normal without external vision systems. The control framework, depicted in Fig. 4, addresses three core challenges as follows.

1) *Adaptive Z-Motion:* To ensure uniform imaging across heterogeneous tissues, we propose a fusion filtering strategy. This integrates the CM (as prior knowledge) with force sensor data using Bayesian estimation to dynamically determine the optimal contact force. Crucially, a dual-layer safety mechanism is embedded within this loop, enforcing both a global hard force threshold and an adaptive soft limit to prioritize patient safety.
2) *High-Bandwidth In-Plane Servo:* We introduce a lightweight CS algorithm to calculate $\theta_{in\text{-}plane}$. Unlike pixel-wise CM methods, this approach significantly reduces computational latency, enabling high-frequency real-time tracking.
3) *Biomechanics-Informed Out-of-Plane Strategy:* We employ the Y–L equation to model the epidermal surface and predict $\theta_{out\text{-}of\text{-}plane}$. To address the bandwidth mismatch between the fast in-plane servo and the computationally intensive Y–L fitting, we implement a trajectory planning strategy that minimizes forward curvature. This decouples the orientation adjustments, ensuring the slower out-of-plane controller has sufficient convergence time.

III. Contact Force and Normal Searching in In-Plane Orientation

Stable imaging quality and precise in-plane servo control are essential for autonomous US scanning. The variability in anatomical structures across patients means that employing a constant contact force often results in inconsistent imaging quality. To achieve stable imaging quality across different areas and individuals, this chapter proposes a Z-motion control method based on fusion filtering. In addition, the study proposes a novel in-plane servo algorithm that utilizes CS to ensure both high accuracy and high bandwidth for in-plane motion.

A. Z Motion Control in US Scanning

As established in Section II-A, the ultimate goal of z-motion control is to achieve and maintain stable, high-quality imaging. Conventional RUSSs often employ a constant-force control strategy, which assumes that a stable contact force ensures consistent image quality. However, this assumption does not hold in practice due to the dynamic nature of biological tissues. Our approach departs from this notion by designing a closed-loop controller that uses contact force merely as a control variable, while the actual target is to stabilize image quality itself. Through real-time Bayesian fusion and adaptive motion regulation, the system is capable of compensating for anatomical variability during scanning.

Physical metrics like SNR are impractical in this context due to nonstandard anatomical structures, dynamic motion precluding temporal analysis, and the unavailability of ROIs during the search. Instead, we employ a CM as a task-oriented evaluator of anatomical recognizability. Its zero-order moment quantifies semantic presence, serving as the high-level prior $P(\theta)$ in our probabilistic model.

To combine this perceptual prior with physical sensing, Bayesian estimation is employed to fuse the confidence-map information with force sensor data, formulated as

$$P(\theta \mid X) = \frac{P(X \mid \theta)\ P(\theta)}{\int_{\Theta} P(X \mid \theta)\ P(\theta)\ d\theta}. \tag{4}$$

Given a dataset X with each measurement $x_i \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and a prior $\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$, the posterior is

$$P(\theta|X) = \frac{1}{\sqrt{2\pi}\sigma_N} \exp\left[-\frac{1}{2}\frac{(\mu - \mu_N)^2}{\sigma_N^2}\right] \to \mathcal{N}\left(\mu_N, \sigma_N^2\right) \tag{5}$$

$$\mu_N = \frac{N\sigma_0^2}{N\sigma_0^2 + \sigma_x^2}\frac{1}{N}\sum_{i=1}^{N} x_i + \frac{\sigma_x^2}{N\sigma_0^2 + \sigma_x^2}\mu_0$$

$$\sigma_N^2 = \frac{\sigma_0^2\sigma_x^2}{N\sigma_0^2 + \sigma_x^2}. \tag{6}$$

The updated mean μ_N represents the dynamic optimal contact force F_m, which achieves the desired image quality

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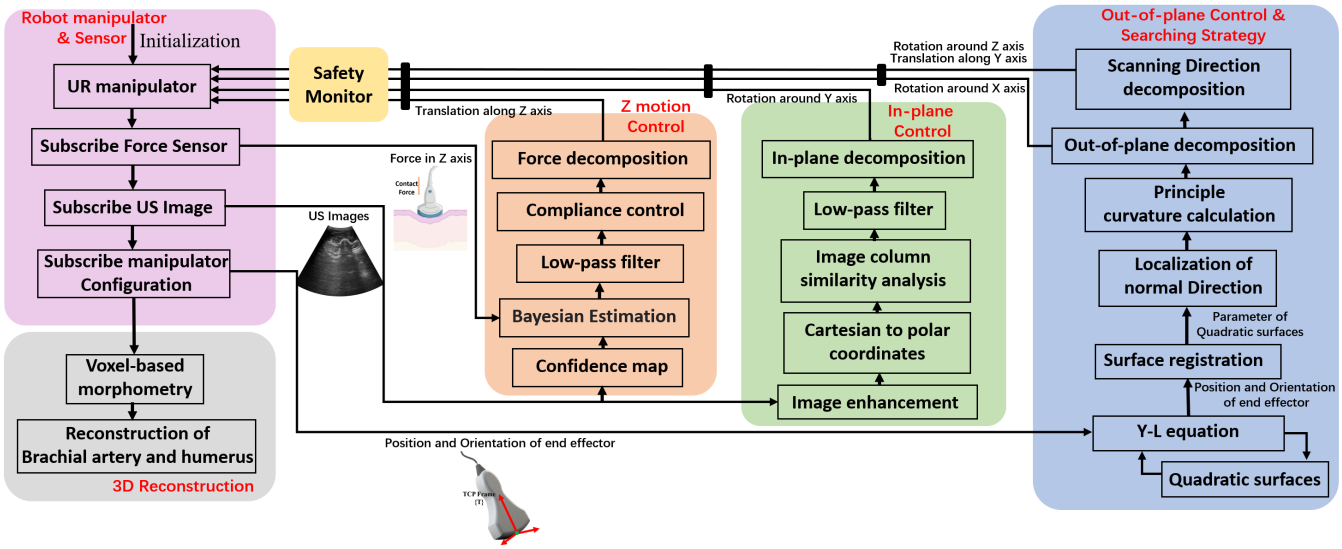


Fig. 4. Algorithmic diagram of the US autonomous scanning system targeting an unknown epidermal surface.

under current anatomical conditions. A low-level admittance controller then enforces this force target compliantly along the Z-axis

$$k_m\ddot{x}_e + k_d\dot{x}_e + k_s x_e = F_m - F_{ext}. \tag{7}$$

Here, k_m, k_d, and k_s denote the virtual mass, damping, and stiffness; x_e is the probe displacement; and F_{ext} is the externally measured contact force. The term $(F_m - F_{ext})$ drives compliant motion to realize smooth force adaptation and minimize tissue deformation.

To ensure that this responsive control remains both safe and comfortable for patients, a dual-layer intelligent safety mechanism is further integrated. This mechanism operates in parallel with the control loop, acting as a supervisor to constrain the target force F_m in (7).

1) Layer 1 (Global Hard Safety Threshold): A global upper bound of 25 N is conservatively set according to human-subject testing. This hard safety limit acts as a fail-safe condition: once exceeded, all motion commands are immediately suspended to prevent any potential harm.

2) Layer 2 (Adaptive Soft Force Limit): To maintain comfort and avoid redundant pressing, a soft safety layer continuously monitors the gradient of image quality with respect to force ($dI_{quality}/dF$). When the improvement in quality saturates (i.e., the gradient approaches zero), the algorithm dynamically lowers the operational force ceiling. This effectively clamps the target force F_m, preventing it from increasing further into the zone of diminishing returns. This mechanism enhances both user comfort and scanning efficiency without compromising image integrity.

Together, these two layers form an intelligent safety framework that maintains a balance between robust imaging performance and strict patient protection standards.

B. Extraction of Normal Direction From US Image

To determine the normal direction within the plane, various methods have been developed and explored [17], [19], [22], [24]. The CM method [16] is recognized as the predominant technique in numerous autonomous US scanning systems. However, the CM method necessitates substantial computational resources to compute the random walk probability for each point and is prone to noise interference. To address this issue, the current study presents an innovative computational approach grounded in CS, which ensures enhanced control frequency and augmented accuracy for such tasks.

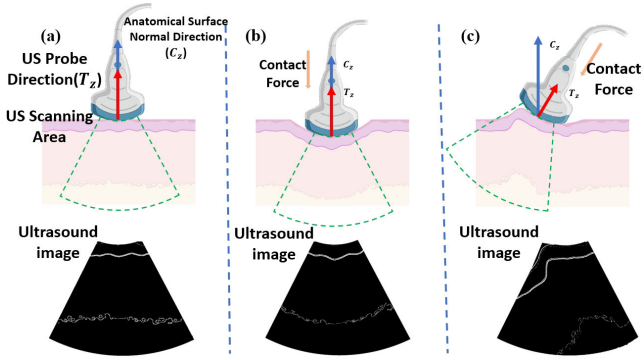


Fig. 5. Impact of tissue-probe interaction modes, including (a) quasi-contact, (b) fully contact, and (c) contact with in-plane error.

The anatomical structures in the human body typically consist of multiple layers, each characterized by distinct acoustic properties, as illustrated in Fig. 5(a). US images often depict these layers as varying degrees of brightness. Under the standard pressure applied by an ultrasound probe—particularly a convex type—these layers tend to develop strains that are concentrated around the probe's contact point. In ultrasound imagery, this effect manifests as bands that appear roughly concentric, as depicted in Fig. 5(b).

However, when the pressure from the US probe is misaligned with the surface normal, the structures do not form concentric bright bands, as depicted in Fig. 5(c). Furthermore, if any part of the US probe's array loses contact with the skin, the corresponding area appears black in the image, indicating an absence of returned signals.

Therefore, when tissue is compressed in alignment with the normal direction, similar reflection waves are generated

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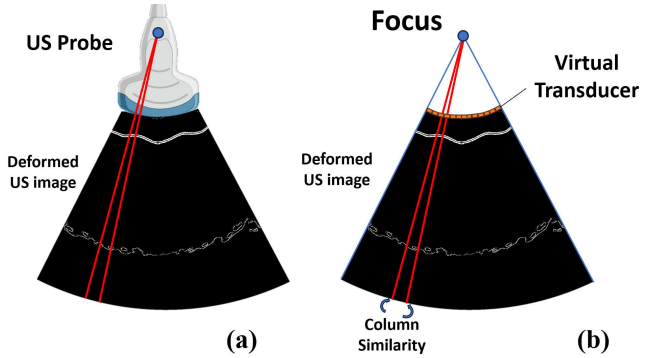


Fig. 6. Virtual transducer and deformed US image. (a) Illustrates the fundamental ultrasound scanning scenario under tissue compression. (b) Schematically details the analytical method based on a virtual transducer and the principle of column similarity. The acoustic waves from individual transducers are assumed to propagate independently along paths defined by a common focus. By analyzing the correlation of echo intensities from adjacent transducers, the tissue's normal direction can be inferred.

by various transducers. This study posits that the US waves produced by each transducer operate without interference. In addition, it is assumed that the sound waves propagate along the paths defined by the focus and each respective transducer. Consequently, the echo signal received by each transducer corresponds to the echo intensity along its respective ray, as illustrated in Fig. 6(b). By analyzing the correlation between echo intensities of adjacent transducers, it is possible to identify changes in anatomical structures, thereby determining the normal direction within the $\theta_{in\text{-}plane}$ orientation.

Algorithm 1 Calculate $\theta_{in\text{-}plane}$ Base on CS

Input: Ultrasound image $\Omega_{original}$
Output: $\theta_{in\text{-}plane}$
if *Probe type* **is** *Convex probe*
  Interpolate Ω According polar coordinate;
$\Omega_{shift} \leftarrow$ Bit cycle shift of $\Omega_{original}$
$mux2 \leftarrow$ Square of average of $\Omega_{original}$ by column
$muy2 \leftarrow$ Square of average of Ω_{shift} by column
$muxy \leftarrow$ Bit multiplication of $\sqrt{mux2}$, $\sqrt{muy2}$
$sigmax2 \leftarrow$ (Average of Ω^2_{shift} by column) - mux2
$sigmay2 \leftarrow$ (Average of Ω^2_{shift} by column) - muy2
$sigmaxy \leftarrow$ (Average of $\Omega_{shift} * \Omega_{shift}$ by column) - muxy
$C(r,\theta) \leftarrow$ Calculate column similarity by $\frac{((2*muxy+C1)*(sigmaxy+C3)*2)}{((mux2+muy2+C1)*(sigmax2+sigmay2+C2))}$
$C_m(\theta) \leftarrow$ Average of $C(r,\theta)$ by column
$C_l(\theta) \leftarrow$ Low-pass filtering on $C_m(\theta)$
$\theta_{in\text{-}plane} \leftarrow \int_\theta \theta C_l(\theta)\, d\theta$
return $\theta_{in\text{-}plane}$

Based on this principle, the study assesses whether the US probe is perpendicular to the skin by evaluating the structural similarity of column pixels, which correspond to each angle in the polar coordinate system. The experiment utilizes a probe equipped with 1024 virtual transducers, each capturing 512 echo signals. Algorithm 1 delineates the pseudocode representation of the method. Initially, the US image is interpolated and converted into polar coordinates. Subsequently, the structural similarity of the image is computed. Finally, by analyzing the distribution of CS, the angle between the probe's normal and the epidermal surface's normal in the in-plane direction is determined. The experimental results remarkably demonstrate the stability of the algorithm. Compared to traditional methods, the proposed method in this article reduces the computing time by 84.6% and decreases the average error by 33.0%. The algorithm has been verified on various phantoms and human body parts, proving its strong robustness.

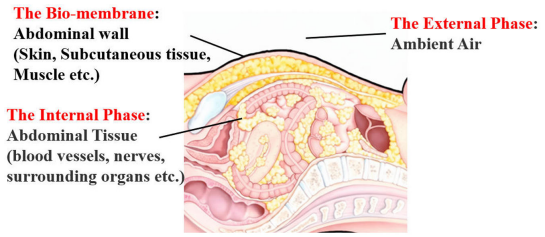


Fig. 7. Conceptual illustration of the mechanical model for the abdominal wall. The system is modeled as a "composite bio-membrane" (representing the abdominal wall) separating an "external medium" (air) from an "internal medium" (an effective fluid representing the abdominal contents).

IV. Normal Detection in Out-of-Plane Orientation and Scanning Strategy

In addition to in-plane motion, controlling the US probe's movement beyond the plane is essential for acquiring volumetric information about the tissue. This section describes the method developed to control the probe's motion beyond the plane, including Y-axis translation T_{Ty}, Z-axis rotation R_{Tz}, and X-axis rotation R_{Tx}, as illustrated in Fig. 3. Among these movements, R_{Tx} corresponds to the out-of-plane rotation, aimed at obtaining a high-quality image. Meanwhile, T_{Ty} and R_{Tz} determine the probe's subsequent direction of movement. This section first introduces the use of the Y–L equation in US scanning. It then derives the local epidermal surface equation based on previous points and a liquid surface model, through which the surface normal and principal directions are determined. Finally, we leverage this information to develop an out-of-plane decomposition and describe subsequent steps in the scanning strategy.

A. Y–L Equation and Surface Representation

To accurately predict and track the dynamically changing body surface during autonomous robotic scanning, we introduce a physics-based surface representation. The core of this method is to model the local body region under probe pressure, such as the abdomen, as a mechanical system whose behavior is governed by the Y–L equation from biomechanics.

The theoretical foundation of this model lies in simplifying the complex biological structure into a three-component mechanical system, as illustrated in Fig. 7. This model comprises: 1) an external medium, which is the ambient air; 2) an interfacial layer, representing the abdominal wall, which we model as a single "composite bio-membrane" that includes the skin, fat, and muscle layers, collectively exhibiting the mechanical properties of an elastic membrane; and 3) an internal medium, consisting of the organs, tissues, and fluids within the abdominal cavity, whose collective mechanical response

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is homogenized into an "effective fluid." This approach of simplifying a pressurized biological tissue into an interface between an elastic membrane and a fluid is a well-established paradigm in fields like cell biomechanics and soft tissue mechanics [30], [31], providing a solid theoretical basis for our model.

A critical consideration is that the Y–L equation describes a static fluid interface, whereas robotic scanning is inherently a dynamic process. We address this by invoking the "quasi-static approximation." The validity of this assumption stems from the significant separation in time scales: the robot's scanning velocity (typically in the order of cm/s) is much slower than the propagation speed of shear waves in soft tissue (typically in the order of m/s). This means that at each instant during the robot's movement, the mechanical state within the tissue can rapidly reach equilibrium. Therefore, at any given moment during the scan, the static Y–L equation can be used to accurately approximate the relationship between the surface morphology and the applied pressure.

The classic Y–L equation relates the pressure difference Δp across a static fluid interface to the interfacial tension and principal curvatures

$$\gamma\left(\frac{1}{R_1}+\frac{1}{R_2}\right)-\rho g z=\Delta p \tag{8}$$

where γ is the surface tension coefficient, R_1 and R_2 are the principal radii of curvature at a point on the surface, ρ is the fluid density, g is the gravitational acceleration, and z is the height. In our application, the pressure changes induced by the probe's contact force are significantly larger than the minute pressure gradients caused by gravity (the $\rho g z$ term). Consequently, this gravitational term can be safely neglected, simplifying the equation to a direct relationship between the pressure difference and the mean curvature.

However, solving this nonlinear partial differential equation in real-time is highly challenging, especially in the presence of physiological disturbances like respiration and cardiac motion. To achieve robust and efficient surface fitting, we locally approximate the body surface with a quadratic function and employ an optimization framework that integrates geometric information with the physical constraint. Specifically, we use the previously acquired waypoints $P_i(x_i, y_i, z_i)$ and their corresponding tangent directions $\vec{d}_i$ (obtained from the in-plane servoing) as geometric priors. The surface fitting is then formulated as the following nonlinear least-squares optimization problem:

$$\begin{aligned}\min_{a,b,c,d,e,f,\{\Delta p_i\}} & \sum_{i=1}^{n}\left(z_i-\left(ax_i^2+by_i^2\right.\right.\\ &\left.\left.+cx_iy_i+dx_i+ey_i+f\right)\right)^2\\ &+w_t\sum_{i=1}^{n}\left(\nabla S\left(x_i,y_i\right)\cdot\vec{d}_i\right)^2\\ &+w_p\sum_{i=1}^{n}\left(\gamma\left(\frac{1}{R_{i1}}+\frac{1}{R_{i2}}\right)-\Delta p_i\right)^2\end{aligned} \tag{9}$$

where $S(x,y)=ax^2+by^2+cxy+dx+ey+f$ is the quadratic surface to be determined. The objective function comprises three terms: the first is a geometric position fitting term, ensuring the surface passes through the historical waypoints; the second is a geometric tangent constraint, enforcing that the surface gradient aligns with the known probe tangent direction; and the third is the physical constraint term, ensuring the fit surface's shape (represented by its principal curvatures R_{i1}, R_{i2}) adheres to the Y–L equation. The coefficients w_t and w_p are weights that balance these terms. We use the 50 most recent waypoints, remove outliers using the RANSAC algorithm, and solve this optimization problem via nonlinear least squares.

Especially, when the probe moves to regions where thin soft tissue directly overlays bone (e.g., the clavicle apex or ribs), the "effective fluid" assumption weakens. In this case, the framework does not require an explicit fallback strategy. The optimization in (9) contains three terms; when the Y–L physical constraint becomes less representative, its residual grows, and the two geometric terms (position fitting and tangent constraint) naturally dominate the solution. The Y–L term thus degrades to a smoothness regularizer rather than a strict physical constraint. In addition, RANSAC treats sharp curvature fluctuations from underlying bone as outliers, and the sliding window of 50 waypoints ensures that abrupt tissue-to-bone boundaries are absorbed gradually rather than encountered as discontinuities.

As demonstrated by the experiments in Section V, this method accurately and robustly reconstructs the local convex structures of the body surface.

B. Localization of Normal Direction and Principle Curvature Calculation

To estimate the out-of-plane orientation and ascertain the scanning direction for the US probe at the surface coordinate $P_0=(x_0,y_0,z_0)$ (which is the next scanning point), it is crucial to determine the normal and principal directions. We first calculate the normal vector $\vec{\mathbf{N}}$ of the quadratic surface at next scanning point P_0

$$\vec{\mathbf{N}}=\left(\frac{\partial z}{\partial x},\frac{\partial z}{\partial y},-1\right)=(2ax_0+cy_0+d,2by_0+cx_0+e,-1) \tag{10}$$

where a, b, c, d, e denote the parameters of the quadratic surface, while $\vec{\mathbf{N}}$ represents the normal vector at that point. To further determine the principal curvature directions, we next compute the Hessian matrix of the quadratic surface

$$H=\begin{pmatrix}\frac{\partial^2 z}{\partial x^2} & \frac{\partial^2 z}{\partial x\partial y}\\ \frac{\partial^2 z}{\partial y\partial x} & \frac{\partial^2 z}{\partial y^2}\end{pmatrix}=\begin{pmatrix}2a & c\\ c & 2b\end{pmatrix}. \tag{11}$$

Therefore, by solving the eigenvectors $v_i=(v_{ix},v_{iy})$ of the Hessian matrix H, the principal curvature directions $\vec{\mathbf{d}}_{R_1}$ and $\vec{\mathbf{d}}_{R_2}$ can be computed as

$$\begin{aligned}\vec{\mathbf{d}}_{R_1}&=\left(v_{1x},v_{1y},-\nabla z_0\cdot\mathbf{v}_1\right)\\ \vec{\mathbf{d}}_{R_2}&=\left(v_{2x},v_{2y},-\nabla z_0\cdot\mathbf{v}_2\right).\end{aligned} \tag{12}$$

In Sections IV and V, this study clarifies the methodology used to determine the scanning direction from the principal directions $\vec{\mathbf{d}}_{R_1}$ and $\vec{\mathbf{d}}_{R_2}$.

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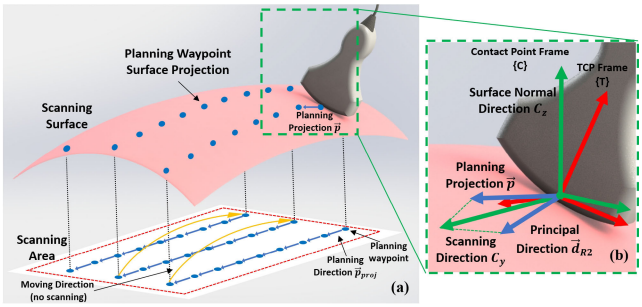


Fig. 8. Ultrasonic autonomous scanning strategy. First, the robot generates waypoints based on the scanning region [red region in (a)]. Next, it determines the scanning direction $\vec{\mathbf{p}}$, then estimates the principal direction $\vec{\mathbf{d}}_{R2}$ based on the Y–L surface, and finally solves for the scanning direction C_y. (a) Generating the scanning direction projection $\vec{\mathbf{p}}$. (b) Solving for the scanning direction C_y.

C. Scanning Strategy and Motion Planning

The algorithm described in Section IV-B involves fitting a surface using a series of in-plane tangents to determine the out-of-plane decomposition and compute the principal directions of the surface. Due to the complexity of the algorithm, the out-of-plane decomposition loop is typically several times slower than the in-plane loop. As a result, the out-of-plane adjustment of the US probe is relatively slow. This performance disparity between the high-bandwidth in-plane controller and the low-bandwidth out-of-plane controller is a critical challenge. A sharp turn in the scanning path would demand a rapid change in the probe's out-of-plane orientation. The slow controller, unable to track this demand, would introduce significant transient error, compromising the probe's normal alignment to the skin and degrading image quality. By proactively minimizing the forward curvature, we generate a smoother, more predictable path. This strategy effectively decouples the fast translational motion from the slow orientational adjustments, granting the out-of-plane controller sufficient time to converge and ensuring stable, high-quality imaging throughout the scan. Therefore, the goal of the scanning strategy is to minimize the curvature along the T_y direction. In addition, the scan path should comprehensively cover the entire volume. Consequently, in this article, C_y, which represents the scanning orientation, is determined using the equation below

$$C_y = \frac{\omega_1 \vec{\mathbf{d}}_{R2} + \omega_2 \vec{\mathbf{p}}}{\|\omega_1 \vec{\mathbf{d}}_{R2} + \omega_2 \vec{\mathbf{p}}\|_2}. \tag{13}$$

In the preceding equation, C_y denotes the y-axis direction in the coordinate system $\{C\}$ of the contact point. The variables ω_1 and ω_2 represent weights, while $\vec{\mathbf{d}}_{R2}$ and $\vec{\mathbf{p}}$ indicate the directions of minimum curvature and the planned offline next scanning direction, respectively. In our implementation, we set the weights to $\omega_1 = 0.3$ and $\omega_2 = 0.7$. The weight $\omega_2 = 0.7$ is determined by a clinical constraint: the full scan must complete within 5 minutes, requiring the path-following term to dominate. The remaining $\omega_1 = 0.3$ provides sufficient authority for local surface orientation adjustment. When $\omega_1 \to 0$, the probe ignores local curvature, degrading image quality; when $\omega_1 \to 1$, the probe over-conforms to local topology and violates the time constraint. The chosen values represent a task-level tradeoff, not an anatomy-dependent parameter. We applied the identical weights to the flat abdominal phantom and the high-curvature clavicle (Section V-F) without any adaptation, and both achieved consistent accuracy, confirming robustness across geometrically distinct regions. Clinically, the most common offline scanning path involves parallel lines [10], typically oriented from top to bottom or from left to right, ensuring comprehensive coverage of the entire body. In this article, our planned paths are aligned along parallel lines from bottom to top, as depicted in Fig. 8. To achieve optimal imaging quality through precise positioning of the US probe, we define $\vec{\mathbf{C}}_z = \vec{\mathbf{N}}$, with C_x derived from the Cartesian coordinate system. The transformations ${}^W_T\mathbf{R}$, ${}^C_W\mathbf{R} \in SO(3)$ denote the rotations from the world frame to the US probe frame and from the world frame to the epidermal surface frame, respectively. The translation from the US probe frame to the epidermal surface frame is represented by

$${}^T_C\mathbf{R} = {}^W_T\mathbf{R}^{-1} \cdot {}^W_C\mathbf{R}. \tag{14}$$

The motion sequence is defined by the control frequencies, arranged in descending order from high to low frequency, which enhances motion precision. Specifically, the control sequence commences with z-axis motion, advances to in-plane motion, proceeds to out-of-plane motion, and culminates in the scanning motion. These actions are executed in the order of T_{Tz}, R_{Ty}, R_{Tx}, R_{Tz} and T_{Ty}. The deduced angles of rotation for the x-axis and z-axis are specified as follows:

$$\begin{aligned} \theta_{Tx} &= \arctan 2\left(-r_{32}, \sqrt{r_{31}^2 + r_{33}^2}\right) \\ \theta_{Tz} &= \arctan 2\left(r_{21}, r_{11}\right). \end{aligned} \tag{15}$$

In the aforementioned equation, r_{32} denotes the (i, j) element of the transformation matrix ${}^T_C\mathbf{R}$, and the translation along the y-axis is defined as 0.2 cm per step. These parameters facilitate the out-of-plane decomposition and, respectively, define the scanning direction for the US probe.

V. Experiments and Results

Several experiments were conducted to validate the system. First, to assess the control method for z-axis motion, experiments were performed on different body regions of 15 volunteers (the BMI ranged from 17.9 to 31.2, including five females and 11 males). This section evaluates the proposed autonomous scanning framework by addressing five critical questions as follows.

1) Does the Bayesian optimization-based force control method ensure consistent imaging quality across heterogeneous tissues?
2) Can the CS-based in-plane servo method achieve high-precision positioning with low latency in an embedded low-power platform?
3) Does the Y–L equation-based out-of-plane servo outperform competing models in ultrasound-guided scanning?
4) Is the proposed framework capable of autonomous exploration on arbitrary unknown surfaces?

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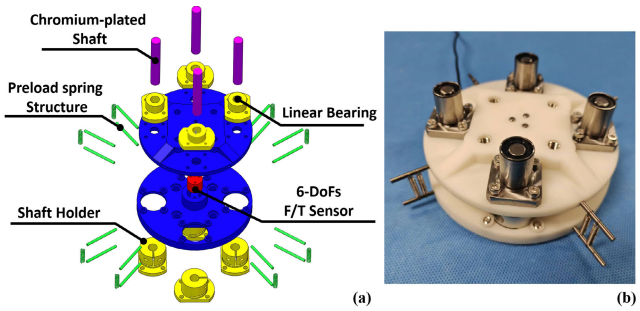


Fig. 9. Supporting structure. (a) Exploded view. (b) Physical diagram.

5) Can the integrated framework autonomously scan human tissues and anatomical structures? Can the reconstruction accuracy of a specific anatomical structure be comparable to that of human experts?

A. System Setup

In our experiment, the UR5 robotic manipulator (Universal Robots, Denmark) was utilized. A 6-axis force/torque sensor, model M3701A (Sunrise Instrument, China), was mounted to the robot's flange via a supporting structure, as shown in Fig. 9. This supporting structure allows for the application of Z-axis pressure on the sensor while limiting forces in other dimensions, ensuring that the force system is exclusively used to maintain contact between the probe and the skin.

In our experimental setup, an Intel Core i5-8400 CPU and an NVIDIA GTX1060 GPU were utilized. The experiments were carried out on a personal computer equipped with 16 GB of memory, serving as the external controller for our robotic system. Ubuntu 20.04 was the operating system of choice, with the Robot Operating System (ROS) enabling seamless communication and control. Evaluation of all controllers was conducted using Python 3.8 and C++. This hardware and software configuration ensured the efficient execution of algorithms and reliable performance throughout our experiments.

To verify the real-time performance of the in-plane method on the embedded low-power platform, a Jetson Orin Nano 8 GB was utilized. Moreover, the algorithm underwent testing with a portable US scanner, model M7, equipped with a sector 2-D US probe 35C50EA (Mindray Corporation, China), noted for its heightened sensitivity in the in-plane direction.

Multimodal data alignment is handled by ROS's message synchronization mechanism, which matches force and image messages by header timestamps with a tolerance of 33 ms (the vision system's update period). At the scanning speed of less than 10 mm/s, a worst case 33-ms mismatch corresponds to 0.33-mm probe displacement, which is below the ultrasound image resolution (0.449 mm/pixel). The temporal mismatch, therefore, falls within subpixel range and does not affect control stability.

B. Dynamic Performance: Constant Force Versus Image Quality Control

To address the need for dynamic performance validation and to clarify our force regulation strategy, we designed an experiment to directly compare the conventional constant force control (Strategy A) against our proposed fusion image quality control (Strategy B). The experiment was conducted on various body parts of a volunteer to assess the strategies' robustness across different tissue stiffness (anisotropy) and under physiological motion.

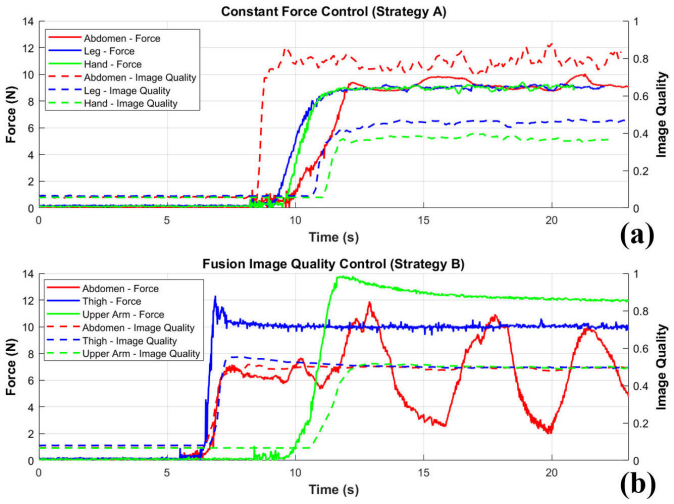


Fig. 10. Comparison of control strategies on different body parts. (a) Strategy A (constant force control) maintains a stable force but results in highly variable image quality due to the tissue's different stiffness. (b) Strategy B (fusion image quality control) maintains a stable, high image quality by dynamically adjusting the contact force. The periodic force modulation on the abdomen compensates for breathing motion. Note the periodic force modulation on the abdomen (green line), which actively compensates for respiratory motion.



Fig. 11. Representative ultrasound frames from the abdominal scan under Strategy A and Strategy B. Top row: With constant force control, the abdominal aorta is intermittently lost, indicating diagnostic unreliability. Bottom row: With our fusion image quality control, the aorta is consistently identified in all frames, ensuring continuous diagnostic value.

Strategy A implements a constant force objective, a common approach in robotic ultrasound. As shown in Fig. 10(a), when a target force of 9N was applied, the system successfully maintained a stable contact force (solid lines) on the abdomen, leg, and hand. However, the resulting IQS (dashed lines) varied significantly. While acceptable quality was achieved on the compliant abdomen, it was substantially lower on the leg and poor on the bony hand, demonstrating that a fixed force does not guarantee consistent image quality across anatomically diverse regions. More critically, during a continuous scan of the abdomen, this strategy's inability to adapt to physiological

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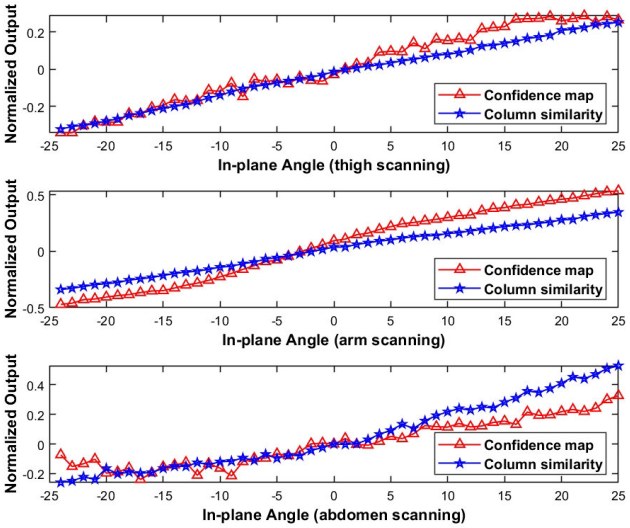


Fig. 12. Comparative experiments of in-plane normal search algorithms. The experiments include scanning of the thigh, upper arm, and abdomen.

motions like breathing resulted in intermittent loss of the target structure. Fig. 11 (top row) provides representative frames where the abdominal aorta, a key diagnostic target, could not be identified at certain moments (e.g., IQS = 0.366), rendering the scan diagnostically unreliable.

In contrast, Strategy B utilizes our proposed method, which dynamically adjusts the contact force to maintain a constant, high level of image quality. The results are presented in Fig. 10(b). The system successfully kept the IQS (dashed lines) stable at a high value (approx. 0.5) on the abdomen, thigh, and upper arm. To achieve this, the contact force (solid lines) became a dynamic variable. Notably, for the abdominal scan, the force exhibited periodic fluctuations, which actively compensated for the tissue deformation caused by the volunteer's breathing. This adaptive behavior ensures that the probe maintains optimal acoustic coupling throughout the respiratory cycle. Crucially, despite these dynamic adjustments, the applied force remained well within the global safety threshold (25 N) defined in Section III-A, ensuring patient safety. The practical benefit of this approach is clearly illustrated in Fig. 11 (bottom row). Throughout the scan, the abdominal aorta remained clearly identifiable in all frames, ensuring the continuous acquisition of diagnostically valuable images.

This comparative experiment unequivocally demonstrates the superiority of our fusion image quality control strategy. By closing the loop on image quality rather than force, our system can intelligently adapt to both static anatomical differences and dynamic physiological disturbances, ensuring consistent and reliable diagnostic imaging, which is a critical failure point for conventional constant-force methods.

C. Experiment of Normal Detection in In-Plane Orientation

1) Performance Comparison With Traditional Methods: To evaluate the effectiveness of the in-plane method, this experiment examined two algorithms: the CM and the CS. This experiment focused on in-plane angle recognition and aimed to assess the algorithms' recognition speed and accuracy across various angles. In the scanning experiment, the robotic arm was programmed to move from an in-plane direction from -25° to $+25^\circ$, performing scans on the human abdomen, upper arm, and thigh. During the scans of the abdomen and thigh, parts of the probe array detached from the epidermal surface at approximately $\pm 19^\circ$ and $\pm 15^\circ$, respectively. The US probe acquired images at a frequency of 30 Hz, which were subsequently stored for algorithmic processing. The experimental results are depicted in Fig. 12. The temporal correlation of the results for the two algorithms during the abdominal, upper arm, and thigh scans yielded coefficients of (0.9976, 0.9932), (0.9982, 0.9891), and (0.9787, 0.9501), respectively. Greater linearity correlates with enhanced detection accuracy and robustness.

The processing speed determines the execution time of the algorithm. This experiment assesses the algorithm's complexity and execution speed by calculating the floating-point operations (FLOPs) and computation time. As shown in Table I, for a single 640×480 image acquired by the US probe, the FLOPs for the CM method and the method proposed in this article are 1.15E+9 and 1.56E+7, respectively. In the experimental environment described, the average processing frequencies are 4.7 and 29.7 Hz, respectively.

2) Real-Time Challenge and GPU Acceleration: Despite the significant speed advantage of a CPU, the initial processing latency of 33 ms (corresponding to a frame rate of 32 Hz) still falls short of the stringent requirements for real-time clinical ultrasound guidance. Standard medical ultrasound machines and their data acquisition cards typically output image streams at 60 Hz. To ensure the algorithm can process every frame without delay, thereby preventing visual stutter and artifacts caused by data loss or processing lag, the processing speed must match or exceed the input frame rate. Therefore, achieving 60 Hz (i.e., a per-frame processing time of less than 16.7 ms) is the critical threshold for providing smooth and reliable real-time visual feedback.

This performance gap motivated us to develop a GPU-accelerated version to overcome the computational bottleneck. As shown in Table II, by migrating the computationally intensive parts of the algorithm (such as the polar transformation and the core similarity computation) to the GPU for parallel processing, the algorithm's latency on a desktop PC (GPU: NVIDIA GTX 1060) was drastically reduced to 3.5 ms, corresponding to a frame rate of 285 Hz, which far exceeds the real-time requirement.

3) Embedded Platform Deployment and Robustness Validation: To further validate the algorithm's practical deployment potential on portable robotic or medical devices, we successfully ported it to the resource-constrained NVIDIA Jetson AGX Orin embedded platform. The results demonstrate that even with significantly lower power consumption and computational power compared to a desktop PC, the algorithm can still operate stably with an average latency of 15.1 ms, corresponding to a frame rate of 66 Hz. This successfully crosses the 60 Hz real-time threshold, proving its feasibility in real-world applications. The experiment results are shown in Table III.

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TABLE I
COMPARATIVE EXPERIMENT OF IN-PLANE NORMAL SEARCH ALGORITHMS

	Scan part	Floating-point operations	Runtime	Mean error	Linearity
Column similarity method	Upper arm	1.5615 E+7	**0.034s ± 0.003s**	**2.93° ± 0.35°**	**0.9976 ± 0.0082**
Confidence map method		1.1573 E+9	0.211s ± 0.015s	4.12° ± 1.13°	0.9932 ± 0.0132
Column similarity method	Thigh	1.5615 E+7	**0.029s ± 0.003s**	**3.04° ± 0.37°**	**0.9982 ± 0.0059**
Confidence map method		1.1573 E+9	0.201s ± 0.011s	3.35° ± 0.49°	0.9891 ± 0.0087
Column similarity method	Abdomen (liver & kidney)	1.5615 E+7	**0.033s ± 0.002s**	**4.12° ± 2.02°**	**0.9787 ± 0.0372**
Confidence map method		1.1573 E+9	0.209s ± 0.013s	6.51° ± 3.15°	0.9501 ± 0.0538

TABLE II
PERFORMANCE COMPARISON BETWEEN CPU-BASED AND GPU-ACCELERATED METHODS

Performance Metric	Original Method (CPU-Based)	Optimized Method (GPU-Accelerated)
I. Overall Performance		
Processing Latency (ms)	35.6 ± 18.4	3.5 ± 0.4
Equivalent Fps (Hz)	29.7 ± 14.1	285.4 ± 27.1
II. Latency Breakdown by Stage		
A. Ingress & Pre-Processing		
Ros Deserialization	1.5 ± 1.2	0.1 ± 0.0
Grayscale Conversion	5.7 ± 8.2	N/A
Polar Transformation	12.4 ± 10.2	1.0 ± 0.4
B. Core SSIM Computation		
CPU→GPU Transfer	N/A	0.2 ± 0.0
Kernel Execution	13.4 ± 5.7 (CPU)	1.6 ± 0.1 (GPU)
GPU→CPU Transfer	N/A	0.0 ± 0.0
C. Egress & Post-Processing		
Egress & Post-Processing	2.4 ± 2.5	0.5 ± 0.1
III. Notes on Implementation		
Core Computation	Numpy on CPU	Cupy on GPU
Polar Transformation	Warppolar + Crop	Pre-Calculated Map

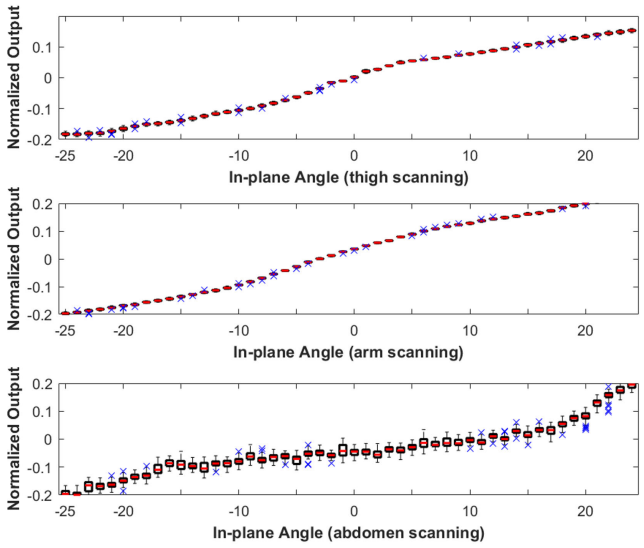


Fig. 13. In-plane tracking performance in multiple BMI volunteers across three distinct anatomical locations: the thigh, arm, and abdomen.

4) Validation on Multivolunteers: Finally, we assessed the algorithm's robustness in challenging clinical scenarios. We specifically selected ultrasound sequences from volunteers (BMI from 16.9 to 31) for testing, as their thicker subcutaneous fat layers often lead to a lower SNR in images, posing a significant challenge for tracking algorithms. The box plot in Fig. 13 shows that there was no statistically significant difference in tracking accuracy or processing speed for the obese patient group. This strongly demonstrates that our method possesses excellent generalization ability and stability across different body types and image qualities.

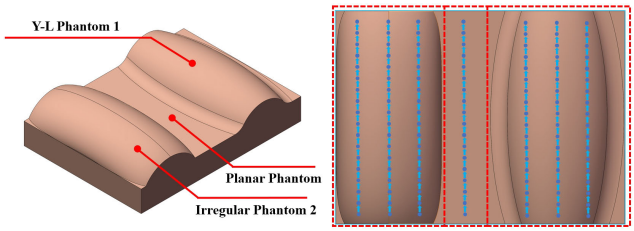


Fig. 14. Comparison of phantom surface models and scanning paths. The left panel illustrates the phantom surface model featuring three distinct surface structures. The right panel displays the offline planned scanning path.

D. Irregular Phantom Scanning

While the ultimate application of our system is on human subjects, the nonrigid nature, physiological motion, and inherent lack of a ground truth for human tissues preclude a rigorous and repeatable quantitative assessment of the control framework's performance, particularly concerning surface reconstruction accuracy and normal alignment. To overcome this limitation and enable a controlled, quantitative validation, we designed a sophisticated physical phantom, as depicted in Fig. 14 (left). This phantom is not a simplistic model; it deliberately incorporates three distinct surface structures—a Y–L surface, an irregular convex surface, and a planar region—to emulate the diverse anatomical topographies encountered in clinical practice. The experimental task required the robotic system to autonomously execute a predefined scanning trajectory designed for comprehensive coverage of this complex surface, as shown in Fig. 14 (right).

To holistically evaluate the system's performance, we established a multifaceted evaluation scheme encompassing both geometric accuracy and imaging quality. Geometric performance was quantified by the surface reconstruction accuracy, measured as the root-mean-square error (RMSE) between the estimated surface and the ground-truth model, and the probe-to-surface normal alignment, assessed by the total angular error. Crucially, imaging performance was evaluated using the composite IQS, as defined in Section II-A. This score provides a comprehensive measure by averaging four normalized metrics: the zero-order moment of the US CM (I_{conf}), SNR,

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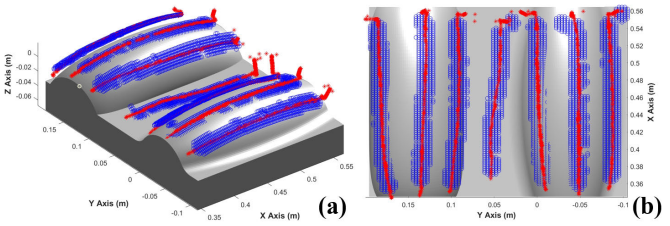


Fig. 15. Visualization of autonomous scanning performance on the irregular phantom. (a) 3-D perspective view showing the executed probe trajectory (red asterisks) closely following the phantom's surface. The blue dots are the surface contact points estimated in real-time by our framework. (b) Top-down view illustrating the high-fidelity tracking of the predefined coverage path. These plots demonstrate the system's capability for simultaneous path following and surface adaptation.

TABLE III

PERFORMANCE COMPARISON BETWEEN PC AND JETSON PLATFORMS

Performance Metric	PC (i5-8400/GTX1060)	JETSON ORIN NANO 8GB
I. Overall Performance		
Processing Latency (ms)	3.5 ± 0.4	15 ± 0.8
Equivalent Fps (Hz)	285.4 ± 27.1	66 ± 4.8
II. Latency Breakdown by Stage		
A. Ingress & Pre-Processing		
Ingress & Pre-Processing	1.1 ± 0.4	3.5 ± 0.4
Ros Deserialization	0.1 ± 0.0	0.3 ± 0.1
Grayscale Conversion	N/A	N/A
Polar Transformation	1.0 ± 0.4	3.2 ± 0.2
B. Core SSIM Computation		
Core SSIM Computation	1.9 ± 0.1	11.1 ± 0.4
CPU→GPU Transfer	0.2 ± 0.0	0.0 ± 0.0
Kernel Execution	1.6 ± 0.1	11.1 ± 0.4
GPU→CPU Transfer	0.0 ± 0.0	0.0 ± 0.0
C. Egress & Post-Processing		
Egress & Post-Processing	0.5 ± 0.1	1.2 ± 0.1

TABLE IV

ANOVA ANALYSIS OF IRREGULAR PHANTOM SCANNING

Metric	F-statistic ($F(df_1, df_2)$)	p-value
Image Quality	F(5, 1540) = 4.84	< 0.001
Total Error	F(5, 1540) = 74.14	< 0.001
In-Plane Error	F(5, 1540) = 88.39	< 0.001
Out-of-Plane Error	F(5, 1540) = 11.56	< 0.001
Sampling Mean Error	F(5, 1540) = 27.60	< 0.001

C_{rel}, and CNR. Finally, to rigorously validate the superiority and necessity of each component in our framework, we conducted comparative experiments against a traditional CM-based approach and ablation studies using simpler surface models (plane and quadratic).

To isolate the variables and rigorously evaluate the intrinsic performance of the six distinct path planning and orientation servoing strategies, the Z-axis controller for this specific experiment was configured to maintain a constant contact force (e.g., approx. 9N). This deliberate design choice deviates from the IQS-stabilizing control mode discussed earlier. By doing so, we ensure that any observed variations in image quality are directly attributable to the efficacy of the path and orientation control method being tested, thus providing an unbiased basis for comparison.

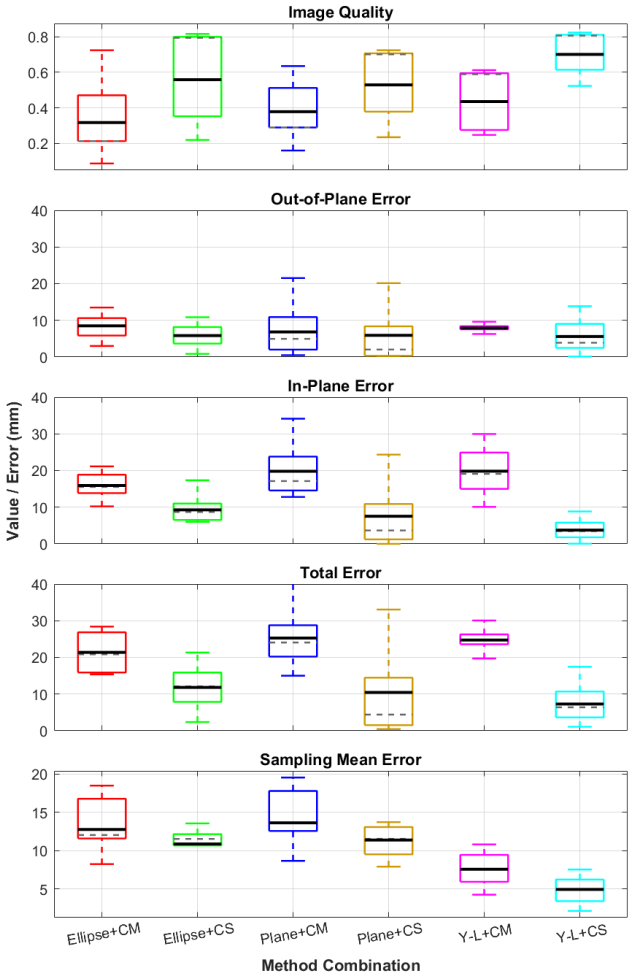


Fig. 16. Comparison of phantom surface models and scanning paths. The left panel illustrates the phantom surface model featuring three distinct surface structures. The right panel displays the offline planned scanning path.

The qualitative results of the phantom scanning experiment are presented in Fig. 15, which visualizes the robot's executed trajectory and the corresponding real-time surface estimation. Fig. 15(a) offers a 3-D perspective, where the red asterisks trace the path of the probe's center-point, and the blue dots represent the surface points estimated by our framework. The tight adherence of the probe's trajectory to the phantom's complex geometry provides compelling evidence of the system's ability to perform real-time adaptation. Specifically, it demonstrates that our visual servoing controller successfully adjusted the probe's height and orientation to conform to the varying curvatures of the Y–L and irregular surfaces. Furthermore, the top-down view in Fig. 15(b) confirms that the system precisely followed the predefined sweeping pattern, ensuring comprehensive coverage of the target areas. The close spatial correspondence between the executed probe locations (red) and the estimated surface points (blue) across the entire scan validates the accuracy and stability of our online surface modeling. Collectively, these results provide strong qualitative validation that our proposed framework can achieve robust and accurate autonomous scanning over challenging, unstructured surfaces, setting the stage for a detailed quantitative analysis.

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TABLE V
COMPARATIVE EXPERIMENTS IN PHANTOM SCANNING

Autonomous Scanning Framework	Imaging Quality	Execution time	Sampling Point Mean Error (Out-of-plane)	Sampling Point Mean Error (In-plane)	Sampling Point Mean Error (Total)	RMSE Around Sampling Point
Sonographer	0.73 ± 0.11	153s	N/A	N/A	$6.98 \pm 2.47^\circ$	$8.65 \pm 1.53mm$
Column Similarity + Y-L equation	$\mathbf{0.70} \pm 0.12$	259s	$\mathbf{5.83 \pm 2.83}^\circ$	$\mathbf{3.78 \pm 2.42}^\circ$	$\mathbf{7.80 \pm 4.40}^\circ$	$\mathbf{4.92 \pm 1.52}mm$
Column Similarity + Plane	0.55 ± 0.26	**243s**	$5.91 \pm 8.00^\circ$	$7.53 \pm 9.43^\circ$	$10.45 \pm 12.47^\circ$	$11.37 \pm 1.88mm$
Column Similarity + quadratic	0.55 ± 0.26	288s	$5.85 \pm 4.20^\circ$	$9.28 \pm 3.12^\circ$	$11.81 \pm 4.85^\circ$	$10.85 \pm 1.56mm$
Confidence Map + Y-L equation	0.43 ± 0.17	357s	$7.91 \pm 1.08^\circ$	$19.83 \pm 5.73^\circ$	$24.72 \pm 3.35^\circ$	$7.56 \pm 2.33mm$
Confidence Map + Plane	0.37 ± 0.18	316s	$6.83 \pm 5.7^\circ$	$19.80 \pm 6.21^\circ$	$25.26 \pm 7.27^\circ$	$13.61 \pm 3.00mm$
Confidence Map + quadratic	0.31 ± 0.24	387s	$8.52 \pm 3.12^\circ$	$15.91 \pm 3.19^\circ$	$21.36 \pm 6.47^\circ$	$12.75 \pm 2.49mm$

Following the qualitative validation, a comprehensive quantitative analysis was conducted to evaluate the performance of our proposed Y–L+CS framework against five alternatives. The descriptive statistics and box plot visualizations for the five key metrics are presented in Fig. 16.

To assess the statistical significance of the observed differences, we first performed a one-way analysis of variance (ANOVA). As summarized in Table IV, the results confirm that highly significant differences exist among the method groups for all five metrics ($p < 0.001$). This foundational finding validates that the choice of method has a profound impact on performance. To identify which specific methods differ, a Tukey's HSD post hoc test was then conducted, with the results indicated by significance brackets in Fig. 16.

The post hoc analysis provides several key insights. First, our Y–L+CS method demonstrated statistically significant superiority in image quality over all five alternatives (pairwise $p < 0.001$). It also achieved the lowest mean values for total error, in-plane error, and sampling mean error, with these improvements being statistically significant against every other method ($p < 0.001$). Furthermore, the analysis of out-of-plane error effectively isolates the contribution of the CS feedback; while statistically comparable to other CS-based methods, Y–L+CS significantly outperformed its direct counterpart Y–L+CM ($p < 0.001$).

Collectively, these quantitative results provide robust statistical evidence that the proposed Y–L+CS framework is the superior approach, excelling in both high-quality imaging and precise geometric control. Results of this experiment is show in Table V.

Quantitative comparison with expert manual scanning highlights the system's precision. The proposed framework achieved a significantly lower RMSE (4.92 ± 1.52 mm) compared to the expert (8.65 ± 1.53 mm), indicating superior geometric reconstruction capabilities. While the expert achieved a marginally higher IQS (0.73 versus 0.70), this was associated with higher and more variable contact forces applied by the human operator. In contrast, the robotic system prioritized a stable, lower force profile to minimize tissue deformation and ensure patient safety, demonstrating a favorable tradeoff between reconstruction accuracy and image quality. The scanning process is as shown in Fig. 17.

Regarding efficiency, the autonomous scan required 259 s, approximately 33% longer than the expert's 153 s. However, this duration reflects the system's strict adherence to the comprehensive path planning strategy, ensuring 100% coverage without the variability inherent in manual operation. Unlike human scanning, which may inadvertently miss regions or vary in speed, the robotic framework guarantees procedural repeatability. This standardized workflow is critical for longitudinal studies and screening applications where consistency outweighs raw speed.

E. Limitation Analysis on a Planar Phantom

To investigate the performance boundaries of our framework, we designed an experiment on a planar phantom. The plane represents a degenerate geometric case, as its normal curvature is theoretically zero in all directions. This poses a challenge for our out-of-plane servo method, which is grounded in the Y–L equation. The Y–L model assumes a nonzero pressure differential (Δp) and, consequently, nonzero principal curvatures. Therefore, this experiment serves as a stress test to evaluate the system's robustness and behavior when operating at the theoretical limits of its underlying biomechanical model.

The plane scanning experiment utilized a phantom system. This system comprises a sound-permeable silicone gel that mimics the acoustic properties and firmness of human skin, an adjustable platform also with sound-permeable properties, and a base, as illustrated in Fig. 18. Specifically, the silicone gel skin features a multilayer structure to accurately emulate the genuine acoustic characteristics and texture of human skin. The platform is perforated with arrays of holes aligned both horizontally and vertically to reflect structures within the human body that exhibit significant acoustic reflectivity. Moreover, the platform's angle can be modified using a screw rod to mimic various surface tilts.

In this experiment, the scanning area was predefined, with the robot autonomously planning waypoints along parallel lines to establish the trajectory, as introduced in Section IV-C. The robotic arm's initial position was assigned randomly. After reaching the first waypoint, the robot executes in-plane movements to explore the normal plane, capturing the point cloud data near the normal, as detailed in Section III-B. Subsequently, the point cloud is aligned to the epidermal surface using the LS method, facilitating out-of-plane control, as shown in Sections IV-A and IV-B. Ultimately, the surface's principal direction is calculated to guide the motion for the next frame, as demonstrated in Section IV-C.

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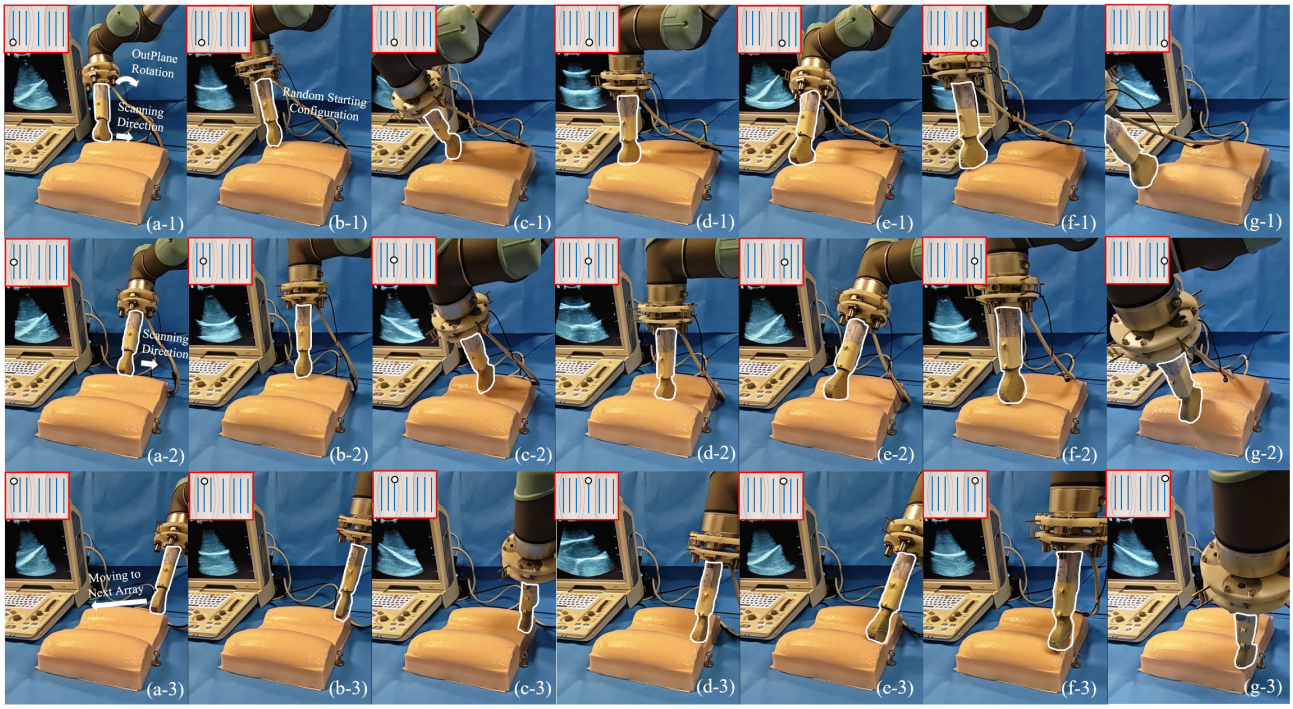


Fig. 17. Results of the phantom scanning experiment. Each column represents a distinct scanning path. The results demonstrate full coverage of the system by the ultrasound scanning frame, and stable tracking of the surface normal by the probe. Each column corresponds to a distinct scanning path on the pahtom. Rows (1)–(3) represent sequential stages of the scanning motion. (a-1)–(g-3) Demonstrate consistent, full coverage of the phantom surface and stable real-time tracking of its normal direction by the robotic probe across all tested paths.

TABLE VI
RESULTS OF PLANE SCANNING EXPERIMENTS

Tilt Angle Degree	Sampling Point Mean Error (Out-of-plane Motion)	Sampling Point Mean Error (In-plane motion)	Sampling Point Mean Error (Total motion)	Mean Error Around Sampling Point
0°	**4.6911°**	4.5616°	7.2315°	7.967°
10°	5.6429°	4.5730°	7.1485°	8.1736°
20°	5.0511°	**4.2165°**	**6.5436°**	**7.4212°**

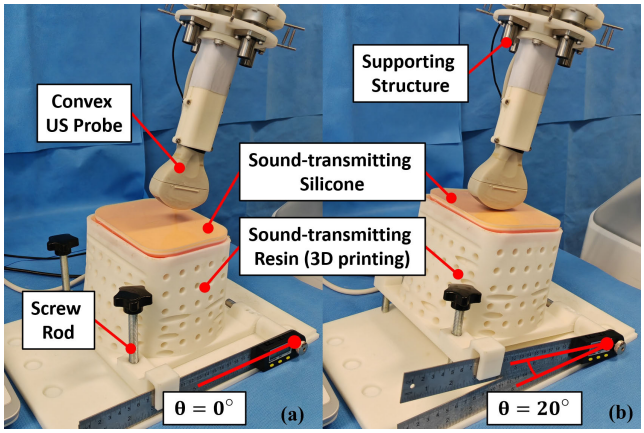


Fig. 18. Plane phantom scanning experimental setup. (a) Ultrasound probe is positioned at $\theta = 0°$ and (b) at $\theta = 20°$. The platform allows for precise control of the phantom's tilt angle to evaluate ultrasound imaging performance under different incident angles.

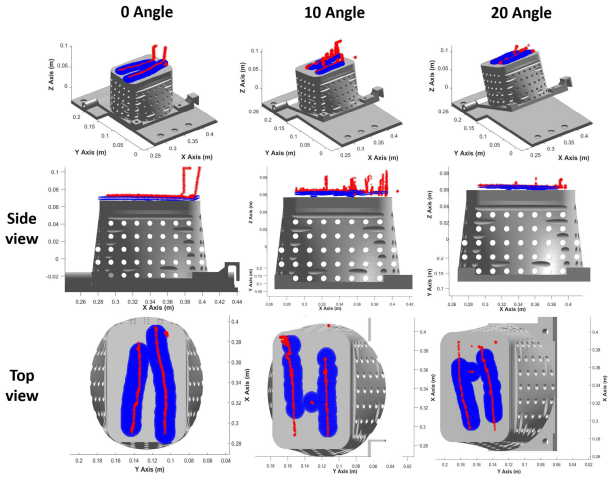


Fig. 19. Plane scanning experiments at various angles. The diagram presents isometric, side, and top views for scans conducted at 0°, 10°, and 20° angles.

The experimental results are depicted in Fig. 19, wherein the system was evaluated at three distinct angles (0°, 10°, and 20°), as depicted in the first row of Fig. 19. Each angle corresponds to a column in the figure, while the three rows represent different viewing angles. The findings reveal that the algorithm's-plane fitting (indicated by the blue spots) and probe location (marked by the red spots) align closely with the

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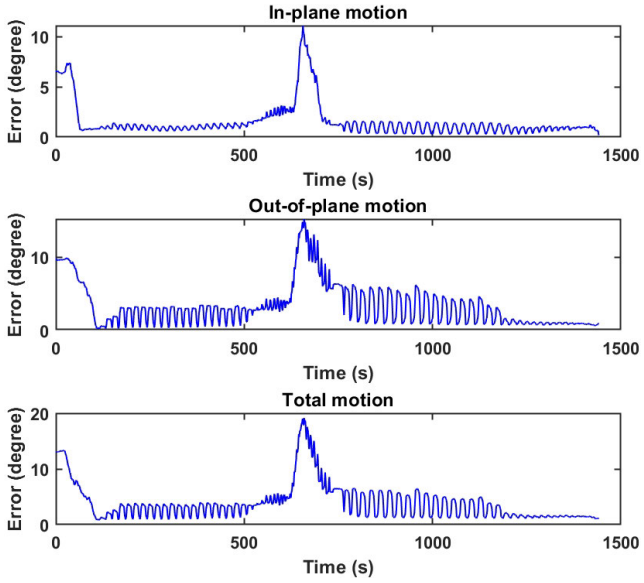


Fig. 20. Angle error analysis for different types of motion. The graph compares the angle errors in in-plane motion, out-of-plane motion, and total motion.

scanned surface across varying viewing angles. Notably, there are abrupt shifts in the in-plane and out-of-plane angles when transitioning between parallel lines, as evidenced in Fig. 20. Moreover, the convergence speed of in-plane, out-of-plane, and total motion progressively diminishes, likely due to the sequentially decreasing control bandwidth. By averaging the fitting surface information at each waypoint, the orientation of the probing plane in different planes can be determined. Table VI presents the in-plane error, out-of-plane error, and total error of the algorithm for each angle. It is crucial to highlight that the algorithm's performance remains largely consistent despite increasing angles. This implies that this method, based on the Y–L equation, is effective for surfaces with a horizontal angle within 20°, meeting the requirements of most scanning tasks conducted in a supine position.

As theoretically anticipated, the scanning trajectory exhibited a minor drift and did not strictly adhere to the preplanned parallel path. This deviation stems from the uniform geometric properties of a plane, which make the determination of a unique principal direction ($\vec{d}_{R2}$) ill-defined. In the absence of a strong curvature gradient, the scanning direction component of the control law [see (12)] is susceptible to small numerical errors. However, it is crucial to note that despite this path drift, the system successfully maintained stable normal alignment throughout the scan, as evidenced by the consistently low total angular errors (below 7.5°) in Table VI. This result demonstrates the framework's robustness: even when operating on a degenerate surface that challenges its path-planning component, the core objective of high-quality image acquisition via normal tracking is preserved.

F. In Vivo Validation and Clinical Feasibility

1) Qualitative Evaluation on Upper Arm: To verify the effectiveness of the algorithm, we conducted a 3-D reconstruction of the humerus in volunteers. In this study, we employ an autonomous scanning method to capture the upper arms of

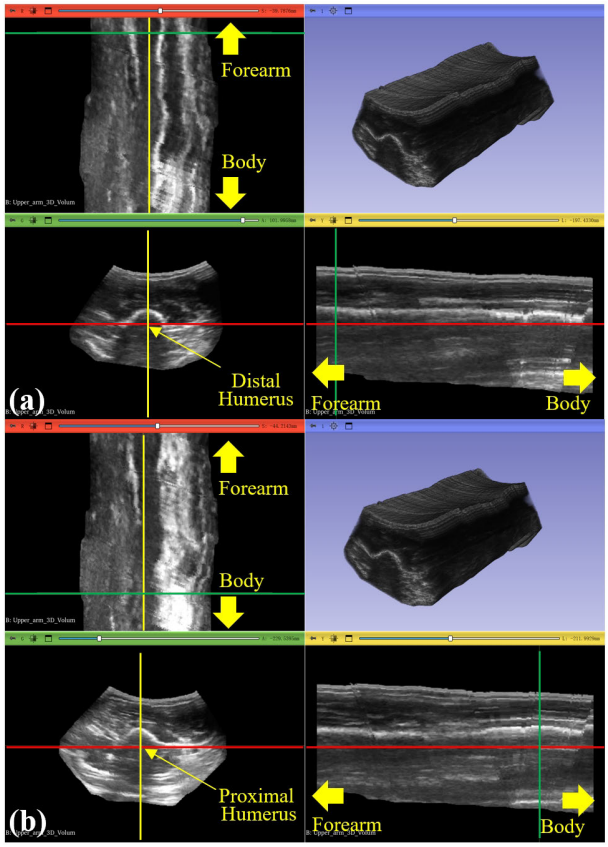


Fig. 21. 3-D volume reconstruction of the proximal humerus and upper arm. The reconstruction includes (a) distal portion of the humerus and (b) proximal portion of the humerus.

volunteers. The control framework ensures physical contact and correct alignment of the robotic probe with the body surface, utilizing a CM for ultrasound image segmentation. Due to the nonrigid nature of the tissues and the lack of a ground truth for evaluation, assessing image quality is challenging. Fig. 21 displays the reconstructed 3-D model of the body. Fig. 21(a) and (b) shows cross-sectional views of the proximal and distal humerus, respectively. The application of 3-D perception allows sonographers to easily identify critical anatomical features, such as the humerus or brachial artery, confirming the clinical diagnostic utility of our algorithm.

2) Geometric Accuracy on Highly Curved Anatomy: To rigorously test the proposed framework's ability to handle challenging geometries, we conducted an in vivo validation study on a highly curved anatomical region—the clavicle of a healthy volunteer. This area is characterized by significant and rapidly changing curvature, presenting a stringent test for the system's surface tracking and normal estimation capabilities. The reconstructed surface was quantitatively compared against a ground-truth model derived from a CT scan of the same subject.

Fig. 22 illustrates the results of this validation. The plot visualizes the robot's TCP trajectory (UR frame points, red), the surface point cloud estimated in real-time by our algorithm (estimated point cloud, blue), and the final reconstructed isosurface. As depicted, the robotic system successfully maintained stable contact and followed the complex contour

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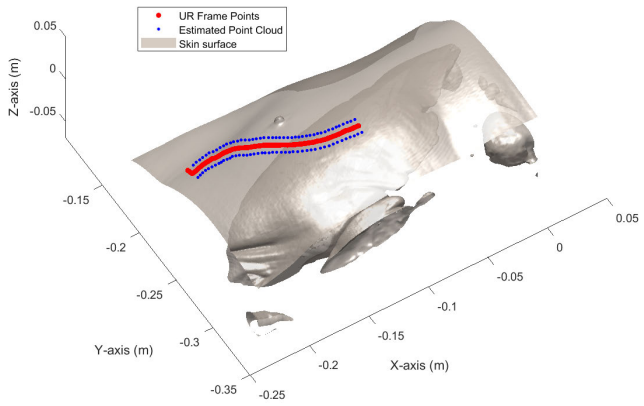


Fig. 22. In vivo validation on the highly curved clavicular region. The system accurately tracks the complex anatomical surface, with the robot's trajectory (red), the algorithm's estimated point cloud (blue), and the final reconstructed surface (gray isosurface) showing strong correspondence.

of the clavicular region, demonstrating the robustness of our Y–L-based normal tracking and scanning strategy.

To quantify the accuracy, the reconstructed surface was compared against a ground-truth model derived from a CT scan of the same subject. The analysis yielded a mean absolute error (MAE) of 6.21 mm and an RMSE of 6.35 mm. The error distribution was consistent, as indicated by a standard deviation (SD) of 1.31 mm, with the maximum observed error being 7.69 mm. These findings validate that our system can reliably operate on and accurately map highly nonplanar biological surfaces, which is a critical capability for scanning many parts of the human body.

3) Clinical Utility in Abdominal Aortic Reconstruction: To demonstrate the clinical potential and diagnostic-level accuracy of our autonomous scanning framework, we performed a complete 3-D reconstruction of the abdominal aorta on a healthy volunteer. This experiment serves as the ultimate validation, assessing the system's performance in a clinically relevant task that requires both stable contact on soft, deformable tissue and the ability to generate high-quality images for accurate modeling. The robotic reconstruction was compared against a 3-D model manually created by an expert sonographer who performed a separate scan on the same subject.

Fig. 23 presents a direct visual comparison of the 3-D models. The model reconstructed by our autonomous robotic system is rendered in red, while the expert's manual reconstruction is shown in green. A high degree of morphological agreement is immediately apparent, particularly in the overall shape, trajectory, and bifurcation of the aorta. This visual evidence suggests that our method can capture the key anatomical features with high fidelity.

For a rigorous quantitative assessment, we evaluated the two models using standard metrics for volumetric overlap and surface distance, and also compared a key clinical parameter.

1) *Volumetric Overlap:* We calculated the DSC to measure the volumetric overlap between the two models. Our framework achieved a DSC of 0.85, indicating excellent volumetric agreement.

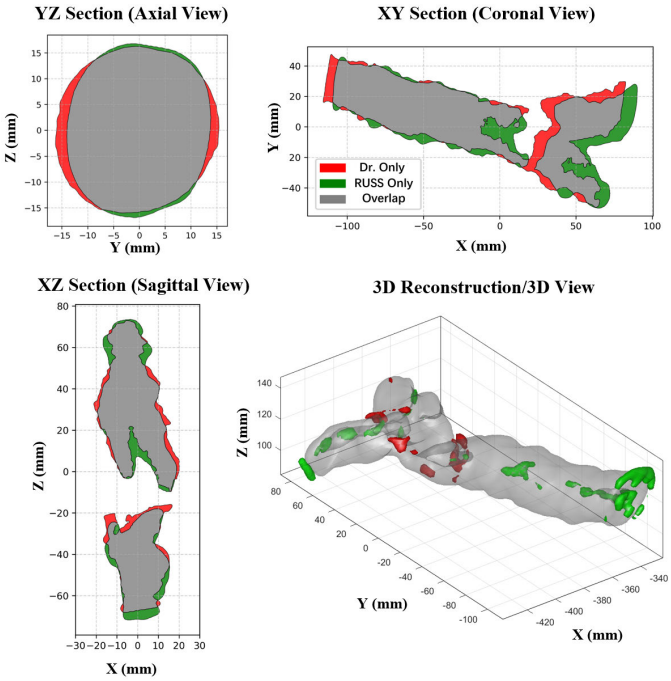


Fig. 23. Comparison of 3-D abdominal aorta models. The reconstruction from our autonomous system (green) shows excellent morphological alignment with the model manually generated by an expert sonographer (red).

2) *Surface Distance:* The average symmetric surface distance (ASSD) was computed to quantify the mean distance between the surfaces of the two models. The resulting ASSD was 2.9 mm, demonstrating high surface reconstruction accuracy.
3) *Measurement of Key Clinical Parameters:* We measured the vessel diameter of the abdominal aorta at three distinct anatomical landmarks (e.g., just below the renal arteries, mid-aorta, and just above the bifurcation). The measurements from our model showed a mean difference of less than 10% compared to the experts' measurements, which is well within the range of clinical acceptability.

Furthermore, the entire autonomous scanning procedure was completed in just 49 s, demonstrating the system's efficiency for clinical workflow. Collectively, these results provide strong evidence for two key conclusions. First, our algorithm is effective for autonomous scanning on the relatively flat but highly deformable abdominal region. Second, the resulting 3-D reconstruction achieves an accuracy, in terms of both geometric fidelity and clinical parameter measurement, that is comparable to the standard of care set by a trained human expert.

This experiment also empirically validates the graceful degradation discussed in Section IV-A. The probe traversed from compliant soft tissue to the bony clavicle apex without parameter adaptation or manual intervention. The low and consistent error (SD = 1.31 mm) across the entire surface indicates no systematic degradation at tissue–bone transition zones, confirming that the geometric terms in the optimization effectively compensated for the weakened Y–L constraint.

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VI. Conclusion

This article presents a control framework for autonomous robotic ultrasound integrating symmetry-optimized image guidance with biomechanics-driven force measurement. By combining a lightweight CS method with a Y–L surface model, the system achieves real-time tracking (66 Hz on NVIDIA Jetson) and elevates the composite IQS by 27.2% compared to conventional methods, with the gap between the proposed method and human expert within 5%. In vivo experiments on complex anatomies, such as the abdominal aorta, confirm that the system achieves volumetric reconstruction accuracy (DSC 0.85) comparable to expert sonographers. While effective for convex probes, the current method relies on specific array geometries. Future work will focus on extending the algorithm's adaptability to linear array probes and integrating semantic understanding for pathological tissue scanning.

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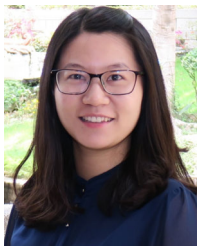
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